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the formation of repeated layers of calcium carbonate makes the hard and glossy pearl. When the pearl enlarges the oyster dies. The shell is then carefully opened and the pearls are manually separated and graded (Figure 13. 12).

Programming of Pearl Industry and Artificial Insertion of Nucleus

This can be achieved by an artificial device to insert the nucleus as foreign particle in the shell of oyster has proved useful for the production of pearls in greater number.

i. Collection of oysters

Oysters are caught by special type of cages (84 x 54 x 20 cm) by covering a heavy wire frame with two centimeter wire mesh. This cage is dipped into a sand-cement mixture providing rough surface to the cages to which free swimming spat get easily stuck up. These cages are suspended at a depth of 6 meters. From July to November, where spats are easily available. These collected oysters are now transferred to rearing cages.

ii. Rearing of oysters

The collected oysters are stocked and reared in special type of cage called as rearing cage. These cages are well protected from enemies of oysters like Octopus, Eel, Devil fishes etc. The collected oysters are first cleaned and then placed into the culture cages for a period of about 10 to 20 days to recover from the strain due to excessive handling and for the physiological adjustment to the shallow water conditions.

iii. Insertion of nucleus

In this method, a piece of mantle of living oyster is cut off and inserted together with a suitable nucleus inside the living tissue of another oyster. Following steps are taken for the insertion of nucleus.

a. Fitness of oyster for operation

The selected oysters for the insertion of nucleus should be healthy and strong enough to overcome the stress during operation

b. Preparation of graft tissues

The piece of tissue which is inserted inside the mantle is called as 'GRAFT' tissue. The outer edges of these graft squares must be known because nacre secreting cells are found only on the outer surface of the mantle so it is essential to keep the outer surface in contact with the inserted nucleus.

c. Preparation of nucleus

Any small particle may function as nucleus to initiate the pearl formation but it is reported that calcareous nucleus is the best because the deposition of nacre was found to be more on calcarious nucleus.

d. Insertion of nucleus

For the insertion of nucleus, oysters are fixed in a desk clamp in the position of right valve facing upward. Mantle folds are smoothly touched to expose the foot and the main body mass, followed by an incision into the epithelium of the foot and a slender channel into the main mass one graft tissue which functions as a bed for the nucleus.

e. Post operation care

Nucleated oysters are placed into cages and suspended into sea water and attached with floating rafts to a depth of 2 to 3 metres for about 6 to 7